

Data Analysis for Business I Midterm 2 Exam

(MG2008)

Total Time (Hrs): 1

Total Marks: 35

Total Questions: 3

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Course Instructor(s)

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Attempt all the questions.

CLO # 2: Compute and interpret various measures of location and measures of variability.

Q1: The following dataset shows the annual revenues (in millions of dollars) for 21 technology companies. [10 marks]

825	12500	2000	4560	3850
12500	498	1670	2300	8200
1460	1910	3400	1100	7100
1800	5100	9500	2740	6200
2650				

- Compute the five-number summary.
- Calculate the lower and upper limits using the interquartile range (IQR) method.
- Identify whether any outliers are present in the data.
- Suppose the revenue value of \$12,500 million was mistakenly entered as \$52,100 million due to a data entry error. Would the outlier detection method used in part (b) help identify this mistake? Justify your answer.
- Draw a box plot to visually represent the data distribution, indicating any outliers, if present.

CLO # 3: Use basic counting techniques, laws of probability, specified discrete and continuous probability distributions in various business problems.

Q2: A recent health and wellness consumer survey was conducted to study preferences for beverages among young adults. Participants were asked whether they preferred herbal tea or an energy drink. A total of 300 individuals participated in the study: 150 men and 150 women. Out of all participants, 210 reported preferring herbal tea and of the group preferring energy drinks, 60 were men and 30 were women. [15 marks]

Let:

M = the event that the participant is a man

W = the event that the participant is a woman

T = the event that the participant prefers herbal tea

E = the event that the participant prefers an energy drink

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- What is the probability that a randomly selected participant prefers herbal tea?
- What is the probability that a randomly selected participant prefers an energy drink?
- What is the conditional probabilities $P(M|E)$ and $P(W|E)$?
- What is the joint probabilities $P(M \cap E)$ and $P(W \cap E)$?
- Given that a participant is a man, what is the probability that he prefers an energy drink?
- Given that a participant is a woman, what is the probability that she prefers an energy drink?
- Is the preference for energy drinks independent of gender? Justify your answer using probabilities.

CLO # 3: Use basic counting techniques, laws of probability, specified discrete and continuous probability distributions in various business problems.

Q3: A mobile app development company launched a new productivity application and created a forecast of potential user growth for the next year based on expert judgment and past trends. The company developed a probability distribution for the number of new downloads in a major city as follows:

Number of Downloads (in thousands)	Probability
50	0.12
100	0.18
150	0.28
200	0.25
250	0.10
300	0.07

[10 marks]

- Is this a valid probability distribution? Justify your answer.
- What is the probability that the app will receive more than 200,000 downloads?
- What is the probability that the app will receive fewer than 100,000 downloads?
- What is the probability that the app will receive at least 50,000 downloads?
- What is the probability that the app will not receive more than 150,000 downloads?